

Chapter 5: screening

Objectives

- *Define screening.*
- *Identify aims and objectives of screening.*
- *Know the differences between screening and diagnostic tests.*
- *Identify types of screening.*
- *Recognize the criteria required to launch a screening program.*
- *Identify measures for evaluation of screening test.*
- *Define and calculate sensitivity, specificity, and yield of the test*

Definition of screening

The search for unrecognized disease or defect by means of rapidly applied tests, examinations or other procedures to sort out apparently well persons who probably have a disease from those who probably do not. Screening is used for case detection, control of disease and research purposes



Fig. (9): Screening

- In passive case finding where patients come to medical personnel with symptoms of disease, they are examined carefully and investigated to be diagnosed.
- On the other hand, in screening, the medical team examines persons without symptoms to detect risk factors or early subclinical disease.

Aims and objectives

- The basic purpose of screening is to sort out from a large group of apparently healthy persons those likely to have the disease or at increased risk of the disease under study.
- Screening is carried out in the hope that earlier diagnosis and subsequent treatment favorably alters the natural history of the disease in a significant proportion of those who are identified as "positive".

▪ Differences between screening and diagnostic tests

▪ Screening test	▪ Diagnostic test
▪ Done on apparently healthy	▪ Done on those with disease indication
▪ Used on groups	▪ Used on individual basis
▪ Less accurate	▪ More accurate
▪ Less expensive	▪ More expensive
▪ Not a basis for treatment	▪ Used as a basis for treatment

Types of screening

1. Mass screening:

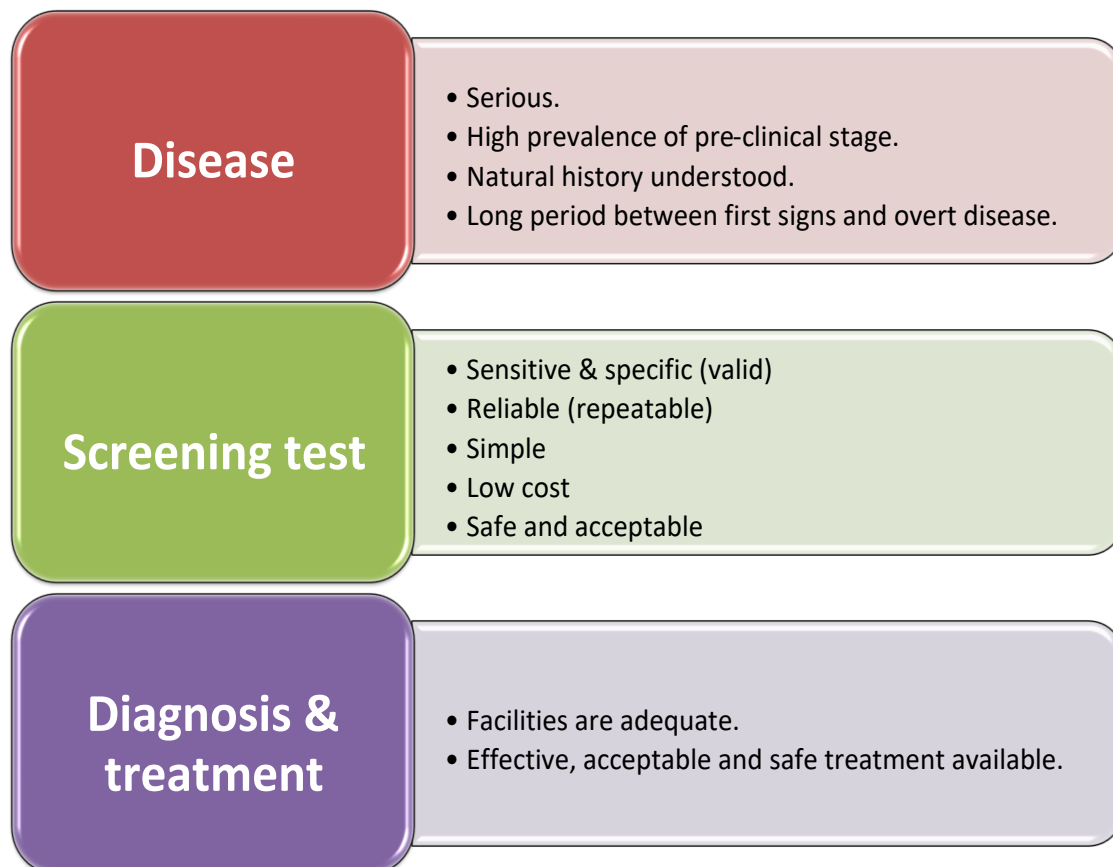
Mass screening simply means the screening of a whole population.

2. High risk or selective screening:

3. Multi-phasic screening:

The application of two or more screening tests in combination to a large number of people at one time.

Criteria for instituting a screening program



Evaluation of screening test

Screening can be evaluated by:

- I. Repeatability (or Reliability) of a screening test.
- II- Validity (or accuracy) of screening test.
- III- Yield of the test.

I. Repeatability (or Reliability) of a screening test.

- A repeatable test is the test that gives the same results if it is repeated many times on the same individual under the same conditions, so repeatability is the percent agreement between multiple results of the test.

Variations in results of the test mean poor repeatability. It may be due to:

- 1- Variation in the characteristic being measured.
- 2- Variation due to observer (tester).
- 3- Variations in instruments or methods.

Measures to enhance repeatability:

- 1- Clear operational definition of variables or cases.
- 2- Good training of data collectors.
- 3- Standardization of questions and tests.
- 4- Data collectors or observers should not be changed in midstream.
- 5- Use of good and well-calibrated instruments.

II- Validity (or accuracy) of screening test

- Validity expresses the ability of a test to separate or distinguish those who have the disease from those who do not.
- It is the degree to which the test measures what is intended to measure.
- It is assessed by comparing the results of the used test with an already accepted test (Valid test).

Sensitivity:

- **The probability of the screening test to be positive when the disease is truly present**

$$\text{Sensitivity} = \frac{\text{True positive (a)}}{\text{True positive (a) + False negative (c)}} \times 100$$

$$\text{Sensitivity} = \frac{\text{Diseased and positive by the screening test}}{\text{Total diseased (validating test)}} \times 100$$

	Disease or Condition	No Disease or Condition
Test Positive	A True Positive	B False Positive
Test Negative	C False Negative	D True Negative

Specificity:

- The probability of the screening test to be negative when the disease is truly absent.

$$\text{Specificity} = \frac{\text{True negative (d)}}{\text{True negative (d) + False positive (c)}} \times 100$$

$$\text{Sensitivity} = \frac{\text{Not diseased and negative by the screening test}}{\text{Total not diseased (validating test)}} \times 100$$

The best screening test is one with high sensitivity and specificity.

Exercise

Valid test screening	<u>ill</u>	Not-ill	
Positive	40	20	60
Negative	60	80	140
	100	100	200

❑ Sensitivity = $40/100 = 40\%$

❑ Specificity = $80/100 = 80\%$

- In a cancer cervix screening program, the screening test was positive in 1300 women and negative in 18700 women. Screening test was positive only in 300 of the 400 confirmed cases.

	ill	Not ill	total
Positive	300	1000	1300
negative	100	18600	18700
total	400	19600	20000

❑ Sensitivity = $300/400 = 75\%$

❑ Specificity = $18600/19600 = 95\%$

III- Yield of the test

- This is the amount of previously unrecognized disease, which is diagnosed and brought to treatment as a result of screening. It depends on sensitivity, specificity and prevalence of the disease.
- The value of screening program is ultimately determined by its effect on morbidity, mortality and disability.

Exercise

1. which of the following is required for enhancing repeatability of the test
 - a) clear operational definition of variables or cases
 - b) variation of the tests used
 - c) changing of data collectors in the mid stream
 - d) using poor calibrated instruments
2. The probability of the screening test to be positive when the diseases is truly present is
 - a) sensitivity
 - b) specificity
 - c) reliability
 - d) yield